

# INTERNATIONAL GCSE **CHEMISTRY**

**9202/1**

Paper 1

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Mark scheme

June 2024

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Version: 1.0 Final



2 4 6 Y 9 2 0 2 / 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from [oxfordaqa.com](https://www.oxfordaqa.com)

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## Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

### Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

### Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

## Information to Examiners

### 1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

### 2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

### 3. Marking points

#### 3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as \* in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

| Student | Response | Marks awarded |
|---------|----------|---------------|
| 1       | green, 5 | 0             |
| 2       | red*, 5  | 1             |
| 3       | red*, 8  | 0             |

Example 2: Name two planets in the solar system.

[2 marks]

| Student | Response                 | Marks awarded |
|---------|--------------------------|---------------|
| 1       | Neptune, Mars, Moon      | 1             |
| 2       | Neptune, Sun, Mars, Moon | 0             |

### 3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

### 3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

### 3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

### 3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

### 3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

### 3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

### 3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

### 3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

### 3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

| Question | Answers     | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------|-------------------|------|-------------------|
| 01.1     | $C_nH_{2n}$ |                   | 1    | AO1<br>3.10.1.3b  |

| Question | Answers     | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------|-------------------|------|-------------------|
| 01.2     | $C_5H_{12}$ |                   | 1    | AO2<br>3.10.1.3ab |

| Question | Answers       | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|---------------|-------------------|------|-------------------|
| 01.3     | decomposition |                   | 1    | AO1<br>3.10.1.3a  |

| Question | Answers  | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|----------|-------------------|------|-------------------|
| 01.4     | monomers |                   | 1    | AO1<br>3.10.2a    |

| Question | Answers                                                                                                                                                                                                                                                                                                                       | Extra information                                                              | Mark | AO/<br>Spec. Ref. |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------|-------------------|
| 01.5     | $  \begin{array}{c}  \text{CH}_3 \quad \text{H} \\    \quad   \\  n \quad \text{C} = \text{C} \\    \quad   \\  \text{H} \quad \text{H}  \end{array}  \longrightarrow  \left( \begin{array}{c} \text{CH}_3 \quad \text{H} \\   \quad   \\ \text{---C---C---} \\   \quad   \\ \text{H} \quad \text{H} \end{array} \right)_n  $ | <p>allow <b>1</b> mark for 5 single bonds</p> <p>allow <b>1</b> mark for n</p> | 2    | AO2<br>3.10.2a    |

| Question    | Answers                                                                                                                       | Extra information                            | Mark | AO/<br>Spec. Ref. |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------|-------------------|
| <b>01.6</b> | any <b>one</b> from: <ul style="list-style-type: none"> <li>not biodegradable</li> <li>not broken down by microbes</li> </ul> | allow does not decompose<br>allow unreactive | 1    | AO1<br>3.10.2e    |

| Question    | Answers                                                                                                                                                 | Extra information                | Mark | AO/<br>Spec. Ref. |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------|-------------------|
| <b>01.7</b> | any <b>one</b> from: <ul style="list-style-type: none"> <li>landfill sites becoming full</li> <li>causes litter</li> <li>harmful to wildlife</li> </ul> | allow more landfill sites needed | 1    | AO1<br>3.10.2e    |

|                         |  |          |
|-------------------------|--|----------|
| <b>Total Question 1</b> |  | <b>8</b> |
|-------------------------|--|----------|

## Question 2

| Question | Answers                          | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|----------------------------------|-------------------|------|-------------------|
| 02.1     | turns damp red litmus paper blue |                   | 1    | AO1<br>3.4.2d     |

| Question | Answers   | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-----------|-------------------|------|-------------------|
| 02.2     | diffusion |                   | 1    | AO1<br>3.1.1b     |

| Question | Answers           | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------------|-------------------|------|-------------------|
| 02.3     | ammonium chloride |                   | 1    | AO3<br>3.4.2d     |

| Question | Answers                                                                              | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|--------------------------------------------------------------------------------------|-------------------|------|-------------------|
| 02.4     | ammonia (molecules) diffuse / travel more quickly (than hydrogen chloride molecules) | allow converse    | 1    | AO2<br>3.1.1b     |

| Question | Answers         | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-----------------|-------------------|------|-------------------|
| 02.5     | OH <sup>-</sup> |                   | 1    | AO1<br>3.5.1df    |

| Question | Answers                               | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|---------------------------------------|-------------------|------|-------------------|
| 02.6     | (mass =) $\frac{10.4}{52} \times 100$ |                   | 1    | AO2<br>3.5.1d     |
|          | = 20 (g)                              |                   | 1    |                   |

| Question    | Answers                                                   | Extra information                                                    | Mark | AO/<br>Spec. Ref.       |
|-------------|-----------------------------------------------------------|----------------------------------------------------------------------|------|-------------------------|
| <b>02.7</b> | (moles = $0.06 \times 8.0$ =) 0.48                        |                                                                      | 1    | AO2<br>3.6.3a<br>3.6.4a |
|             | ( $M_r = 14 + (3 \times 1)$ =) 17                         |                                                                      | 1    |                         |
|             | (mass =) $17 \times 0.48$                                 | allow correct use of incorrectly determined moles and/or $M_r$       | 1    |                         |
|             | = 8.16 (g)                                                |                                                                      | 1    |                         |
|             | <b>alternative method</b>                                 |                                                                      |      |                         |
|             | ( $M_r = 14 + (3 \times 1)$ =) 17 (1)                     |                                                                      |      |                         |
|             | (mass in 1 dm <sup>3</sup> = $0.06 \times 17$ =) 1.02 (1) | allow correct use of incorrectly determined $M_r$                    |      |                         |
|             | (mass in 8 dm <sup>3</sup> =) $1.02 \times 8$ (1)         | allow correct use of incorrectly determined mass in 1dm <sup>3</sup> |      |                         |
|             | = 8.16 (g) (1)                                            |                                                                      |      |                         |

|                         |  |           |
|-------------------------|--|-----------|
| <b>Total Question 2</b> |  | <b>11</b> |
|-------------------------|--|-----------|

**Question 3**

| Question | Answers | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|---------|-------------------|------|-------------------|
| 03.1     | carbon  |                   | 1    | AO1<br>3.2.3b     |

| Question | Answers                                             | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-----------------------------------------------------|-------------------|------|-------------------|
| 03.2     | $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ |                   | 1    | AO2<br>3.6.1a     |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mark | AO/<br>Spec. Ref.            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------|
| 03.3     | <b>Level 2:</b> Scientifically relevant features are identified; the way(s) in which they are similar / different is made clear and (where appropriate) the magnitude of the similarity / difference is noted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4–6  | AO1<br>3.2.1cdefgh<br>3.2.2d |
|          | <b>Level 1:</b> Relevant features are identified and differences noted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1–3  |                              |
|          | <b>No relevant content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0    |                              |
|          | <p><b>Indicative content:</b></p> <p><b>similarities in bonding</b></p> <ul style="list-style-type: none"> <li>• both have strong bonds</li> <li>• have full shells of electrons</li> <li>• bonding involves outer shell electrons</li> </ul> <p><b>differences in bonding</b></p> <p><b>chlorine:</b></p> <ul style="list-style-type: none"> <li>• covalent bonding</li> <li>• (chlorine atoms) share electrons</li> <li>• chlorine molecules</li> <li>• consists of non-metal chlorine atoms</li> <li>• neutral or no overall charge</li> </ul> <p><b>sodium chloride:</b></p> <ul style="list-style-type: none"> <li>• ionic bonding</li> <li>• transfer of electrons</li> <li>• ionic lattice</li> <li>• consists of a metal and a non-metal</li> </ul> <p><b>differences in structure</b></p> <p><b>chlorine:</b></p> <ul style="list-style-type: none"> <li>• simple molecule</li> <li>• intermolecular forces</li> <li>• intermolecular forces are weak</li> <li>• intermolecular forces are random</li> </ul> <p><b>sodium chloride:</b></p> <ul style="list-style-type: none"> <li>• giant structure or lattice</li> <li>• electrostatic forces (between ions)</li> <li>• electrostatic forces are strong</li> <li>• electrostatic forces act in all directions</li> </ul> <p>to access level 2 there must be a comparison of either the structure <b>or</b> bonding of both sodium chloride <b>and</b> chlorine plus a reference to magnitude must also be made.</p> |      |                              |

|                         |  |          |
|-------------------------|--|----------|
| <b>Total Question 3</b> |  | <b>8</b> |
|-------------------------|--|----------|

**Question 4**

| <b>Question</b> | <b>Answers</b> | <b>Extra information</b> | <b>Mark</b> | <b>AO/<br/>Spec. Ref.</b> |
|-----------------|----------------|--------------------------|-------------|---------------------------|
| <b>04.1</b>     | electrons      |                          | 1           | AO1<br>3.2.2g             |

| <b>Question</b> | <b>Answers</b>           | <b>Extra information</b>                     | <b>Mark</b> | <b>AO/<br/>Spec. Ref.</b> |
|-----------------|--------------------------|----------------------------------------------|-------------|---------------------------|
| <b>04.2</b>     | layers (of atoms / ions) |                                              | 1           | AO1<br>3.3.1a             |
|                 | slide (over each other)  | allow atoms / ions can slide over each other | 1           |                           |

| <b>Question</b> | <b>Answers</b> | <b>Extra information</b> | <b>Mark</b> | <b>AO/<br/>Spec. Ref.</b> |
|-----------------|----------------|--------------------------|-------------|---------------------------|
| <b>04.3</b>     | alloy          |                          | 1           | AO1<br>3.3.1b             |

| <b>Question</b> | <b>Answers</b>                                                      | <b>Extra information</b>           | <b>Mark</b> | <b>AO/<br/>Spec. Ref.</b> |
|-----------------|---------------------------------------------------------------------|------------------------------------|-------------|---------------------------|
| <b>04.4</b>     | (initially) as percentage (of tin) increases the strength increases | ignore references to melting point | 1           | AO3<br>3.3.1b             |
|                 | until the strength is 41 (arbitrary units)                          | allow until 50 (% of tin)          | 1           |                           |
|                 | (then) the strength decreases                                       |                                    | 1           |                           |

| <b>Question</b> | <b>Answers</b> | <b>Extra information</b> | <b>Mark</b> | <b>AO/<br/>Spec. Ref.</b> |
|-----------------|----------------|--------------------------|-------------|---------------------------|
| <b>04.5</b>     | 328 °C         |                          | 1           | AO3<br>3.3.1b             |

|                         |  |          |
|-------------------------|--|----------|
| <b>Total Question 4</b> |  | <b>8</b> |
|-------------------------|--|----------|

Question 5

| Question    | Answers                  | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|--------------------------|-------------------|------|-------------------|
| <b>05.1</b> | (A) reactants            |                   | 1    | AO1<br>3.9.2d     |
|             | (B) activation energy    |                   | 1    |                   |
|             | (C) progress of reaction |                   | 1    |                   |

| Question                         | Answers                                   | Extra information | Mark | AO/<br>Spec. Ref. |
|----------------------------------|-------------------------------------------|-------------------|------|-------------------|
| <b>05.2</b><br>mark with<br>05.3 | line of best fit between 4 and 10 minutes |                   | 1    | AO2<br>3.9.2c     |
|                                  | extends line back to y-axis               |                   | 1    |                   |

| Question                         | Answers                                            | Extra information                                                                                                               | Mark | AO/<br>Spec. Ref. |
|----------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------|-------------------|
| <b>05.3</b><br>mark with<br>05.2 | (temperatures =)<br>21.5 (°C) <b>and</b> 17.8 (°C) | allow a tolerance of $\pm \frac{1}{2}$ a small square for the maximum temperature                                               | 1    | AO3<br>3.9.2c     |
|                                  | (temperature change =<br>21.5 - 17.8 =) 3.7 (°C)   | allow correct maximum value from drawn line of best fit<br><br>allow correct use of incorrectly determined temperature value(s) | 1    |                   |

| Question    | Answers                                                                           | Extra information                                                                     | Mark | AO/<br>Spec. Ref.  |
|-------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------|--------------------|
| <b>05.4</b> | (initially) the temperature increases                                             |                                                                                       | 1    | AO2 × 2<br>AO3 × 2 |
|             | (because) energy is given out<br>(by the reaction to the solution / surroundings) | allow (because) the reaction is exothermic                                            | 1    | 3.9.1ab<br>3.9.2c  |
|             | (when the reaction is complete) the temperature decreases                         |                                                                                       | 1    |                    |
|             | (as) energy is transferred (from the solution) to the surroundings                | allow (as) energy is lost (from the solution) to the surroundings / atmosphere / room | 1    |                    |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------|-------------------|
| 05.5     | (improvement)<br>repeat the experiment <b>and</b><br>calculate the mean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   | 1    | AO4<br>3.9.2c     |
|          | (reason)<br>to reduce effect of (random)<br>errors<br><br><b>OR</b><br><br>(improvement)<br>use a lid<br><b>or</b><br>insulate the cup (1)<br><br>(reason)<br>to reduce energy loss (1)<br><br><b>OR</b><br><br>(improvement)<br>measure sodium hydroxide<br>solution with a pipette / burette<br>(1)<br><br>(reason)<br>so more accurate volume (1)<br><br><b>OR</b><br><br>(improvement)<br>use a digital thermometer<br><b>or</b><br>use a temperature sensor and<br>data logger (1)<br><br>(reason)<br>to reduce instrument reading<br>error<br><b>or</b><br>so more accurate temperature<br>value (1) |                   | 1    |                   |

| Question    | Answers                                        | Extra information | Mark | AO/<br>Spec. Ref.       |
|-------------|------------------------------------------------|-------------------|------|-------------------------|
| <b>05.6</b> | same amount of energy given out                |                   | 1    | AO2<br>3.9.2c<br>3.6.4a |
|             | (which) is used to heat a smaller total volume |                   | 1    |                         |

|                         |  |           |
|-------------------------|--|-----------|
| <b>Total Question 5</b> |  | <b>15</b> |
|-------------------------|--|-----------|

## Question 6

| Question | Answers                                                                                                                                          | Extra information                                        | Mark | AO/<br>Spec. Ref. |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------|-------------------|
| 06.1     | any <b>one</b> from: <ul style="list-style-type: none"> <li>• electrical wiring</li> <li>• plumbing</li> <li>• pipes</li> <li>• tanks</li> </ul> | allow copper plating<br>allow catalysts<br>allow coinage | 1    | AO1<br>3.3.1c     |

| Question | Answers                        | Extra information                                                                                          | Mark | AO/<br>Spec. Ref. |
|----------|--------------------------------|------------------------------------------------------------------------------------------------------------|------|-------------------|
| 06.2     | oxidation <b>and</b> reduction | allow loss of electrons <b>and</b> gain of electrons<br><br>allow gain of oxygen <b>and</b> loss of oxygen | 1    | AO1<br>3.8.4ac    |

| Question | Answers                                              | Extra information                                            | Mark | AO/<br>Spec. Ref. |
|----------|------------------------------------------------------|--------------------------------------------------------------|------|-------------------|
| 06.3     | carbon is more reactive (than copper)                | allow converse                                               | 1    | AO1<br>3.3.1.1d   |
|          | (so carbon) will displace copper (from copper oxide) | allow (so carbon) will remove the oxygen (from copper oxide) | 1    |                   |

| Question | Answers                                                         | Extra information                                                                                                                       | Mark | AO/<br>Spec. Ref. |
|----------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------|-------------------|
| 06.4     | $\frac{11.8}{(\text{mass of copper oxide})} = \frac{79.9}{100}$ | allow a correctly calculated answer given to 3 significant figures from an incorrect calculation using the values given in the question | 1    | AO2<br>3.3.1.1c   |
|          | (mass of copper oxide =)<br>$\frac{11.8}{79.9} \times 100$      |                                                                                                                                         | 1    |                   |
|          | = 14.768 (g)                                                    |                                                                                                                                         | 1    |                   |
|          | = 14.8 (g)                                                      |                                                                                                                                         | 1    |                   |

| Question | Answers                                                 | Extra information                            | Mark | AO/<br>Spec. Ref. |
|----------|---------------------------------------------------------|----------------------------------------------|------|-------------------|
| 06.5     | to limit the environmental impact of traditional mining | allow high-grade copper ores are running out | 1    | AO1<br>3.3.1.1f   |

| Question | Answers     | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------|-------------------|------|-------------------|
| 06.6     | bioleaching |                   | 1    | AO1<br>3.3.1.1f   |

| Question | Answers                                              | Extra information                                                                                                   | Mark | AO/<br>Spec. Ref. |
|----------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------|-------------------|
| 06.7     | (advantage)<br>(using bacteria) uses less energy     | ignore references to cost<br><br>allow (using bacteria) does not lead to destruction of the landscape (like mining) | 1    | AO2<br>3.3.1.1f   |
|          | (disadvantage)<br>(using bacteria) is a slow process |                                                                                                                     | 1    |                   |

| Question    | Answers                                            | Extra information | Mark | AO/<br>Spec. Ref. |
|-------------|----------------------------------------------------|-------------------|------|-------------------|
| <b>06.8</b> | plants (grow and) absorb copper compounds          |                   | 1    | AO1<br>3.3.1.1f   |
|             | (then) burn plants                                 |                   | 1    |                   |
|             | (to) produce ash (which contains copper compounds) |                   | 1    |                   |

|                         |  |           |
|-------------------------|--|-----------|
| <b>Total Question 6</b> |  | <b>15</b> |
|-------------------------|--|-----------|

Question 7

| Question | Answers     | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------|-------------------|------|-------------------|
| 07.1     | electrolyte |                   | 1    | AO1<br>3.3.2b     |

| Question | Answers                       | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------------------------|-------------------|------|-------------------|
| 07.2     | keep away from (naked) flames |                   | 1    | AO4<br>3.3.2c     |

| Question | Answers                        | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|--------------------------------|-------------------|------|-------------------|
| 07.3     | (propanone) evaporates quickly |                   | 1    | AO4<br>3.3.2c     |

| Question | Answers                                                                                                                                                    | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------|-------------------|
| 07.4     | electrode will be wet<br><b>or</b><br>there will be solution /<br>propanone on the electrode<br><br>(therefore) the mass would be<br>larger so not correct |                   | 1    | AO4<br>3.3.2c     |
|          |                                                                                                                                                            |                   | 1    |                   |

| Question | Answers                                              | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|------------------------------------------------------|-------------------|------|-------------------|
| 07.5     | (time taken =) $\frac{4.65}{0.005}$<br><br>= 930 (s) |                   | 1    | AO2<br>3.3.2c     |
|          |                                                      |                   | 1    |                   |

| Question    | Answers                                                                                     | Extra information                                                                                                                                                                                                          | Mark       | AO/<br>Spec. Ref. |
|-------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------|
| <b>07.6</b> | (appearance) the colour fades<br><br>(reason) there are fewer copper ions (in the solution) | allow copper ions are discharged (from the solution)<br><br>allow the solution becomes less concentrated<br><br>allow copper ions are removed (from the solution)<br><br>allow copper ions are used up (from the solution) | 1<br><br>1 | AO2<br>3.3.2ac    |

| Question    | Answers                                                                                                                           | Extra information | Mark                | AO/<br>Spec. Ref.                |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------|----------------------------------|
| <b>07.7</b> | (gas) hydrogen<br><br>(because) the solution contains hydrogen ions (from the water)<br><br>(hydrogen is) less reactive than zinc |                   | 1<br><br>1<br><br>1 | AO2 × 2<br>AO3 × 1<br>3.3.2adefg |

|                         |  |           |
|-------------------------|--|-----------|
| <b>Total Question 7</b> |  | <b>12</b> |
|-------------------------|--|-----------|

Question 8

| Question | Answers     | Extra information | Mark | AO/<br>Spec. Ref. |
|----------|-------------|-------------------|------|-------------------|
| 08.1     | endothermic |                   | 1    | AO1<br>3.9.1d     |

| Question | Answers                                                           | Extra information                                              | Mark | AO/<br>Spec. Ref. |
|----------|-------------------------------------------------------------------|----------------------------------------------------------------|------|-------------------|
| 08.2     | yield of methanol increases                                       |                                                                | 1    | AO2<br>3.8.2bc    |
|          | (because) equilibrium position shifts in the exothermic direction | allow (because) equilibrium position shifts to give out energy | 1    |                   |

| Question | Answers                                                                                                                                                             | Extra information | Mark | AO/<br>Spec. Ref.             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------|-------------------------------|
| 08.3     | (as pressure increases) the yield of methanol increases                                                                                                             |                   | 1    | AO2 × 2<br>AO3 × 1<br>3.8.2bd |
|          | (as pressure increases) the equilibrium position shifts to the right-hand side<br><b>or</b><br>an increase in pressure favours the forward reaction                 |                   | 1    |                               |
|          | (because) there are fewer moles / molecules (of gas) on the right-hand side<br><b>or</b><br>(because) the forward reaction produces least number of (gas) molecules |                   | 1    |                               |

| Question    | Answers                                                             | Extra information                                                                    | Mark | AO/<br>Spec. Ref.                 |
|-------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|------|-----------------------------------|
| <b>08.4</b> | (number of moles of CH <sub>4</sub> =)<br>$\frac{1440}{24}$<br>= 60 |                                                                                      | 1    | AO2<br>3.6.1d<br>3.6.2a<br>3.6.4d |
|             | (number of moles of CO =<br>60 × 2 =) 120                           | allow correct use of incorrectly<br>determined number of moles of<br>methane         | 1    |                                   |
|             | (mass of carbon monoxide =<br>28 × 120 =) 3360 g                    | allow correct use of incorrectly<br>determined number of moles of<br>carbon monoxide | 1    |                                   |
|             | (mass of carbon monoxide =<br>$\frac{3360}{1000}$ =) 3.36 kg        | allow correct conversion of an<br>incorrect value for mass of<br>carbon monoxide     | 1    |                                   |

| Question    | Answers                                                                                                                                                                                                                                                                              | Extra information                              | Mark | AO/<br>Spec. Ref.            |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------|------------------------------|
| <b>08.5</b> | 6 minutes                                                                                                                                                                                                                                                                            | allow a value in the range 5.8 –<br>6 minutes  | 1    | AO2 × 1<br>AO3 × 1<br>3.8.2a |
|             | any <b>one</b> from:<br><ul style="list-style-type: none"> <li>(at equilibrium) rate of forward<br/>reaction = rate of backward<br/>reaction</li> <li>concentrations (of reactants<br/>and products) remain<br/>constant</li> <li>(after this point) gradient is<br/>zero</li> </ul> | allow (after this point) curve<br>flattens out | 1    |                              |

|                         |  |           |
|-------------------------|--|-----------|
| <b>Total Question 8</b> |  | <b>13</b> |
|-------------------------|--|-----------|